

Criteria for irreducibility of polynomials

Given a domain Z and a polynomial $f(X) \in Z[X]$, we examine various results about its irreducibility.

Reduction modulo a prime ideal

Given I is a prime ideal in $Z[X]$.

Suppose that $f(X) \bmod I$ is monic in $(Z/I)[X]$. A factoring of f yields a factoring of $f(X) \bmod I$. Thus, if $f(X) \bmod I$ is irreducible, then so is $f(X)$.

Quadratic and cubic polynomials

If f has degree 2 or 3, then it has a factor if and only if it has a linear factor.

A linear factor corresponds to a root in $Q = Q(Z)$, the field of fractions of Z . Since this can be infinite, this condition could be difficult to check.

If f is monic, then each factor is monic, so a linear factor corresponds to a root in Z . Moreover, every root is a divisor of $f(0)$. Thus, it may be relatively easy to check that none of these are roots.

Unique factorisation domains

When Z is a UFD, one can use the lemma about content to prove that if f is irreducible in $Z[X]$, then it is irreducible in $Q[X]$ where $Q = Q(Z)$ is the field of fractions of Z . This does not require f to be monic, but to be irreducible in $Z[X]$ we need the content of f to be a unit.

Factorisation of values

When $Z \supset \mathbb{Z}$, a polynomial of degree d is uniquely determined by its values at $d + 1$ points.

Now, if we have a factorisation $f(X) = g(X)h(X)$ and $d = \max\{\deg(g), \deg(h)\}$, then $f(s) = g(s)h(s)$ for each s in Z is a factoring in Z .

We now look at divisors $g(s)$ of $f(s)$ for $s \in S$ where S has at least $d + 1$ elements. Such divisors determine the polynomial g uniquely with coefficients in $Q(Z)$. Similarly, $h(s) = f(s)/g(s)$ determine h uniquely. If the coefficients are not in Z , then this choice of divisors does not correspond to a factoring. Even otherwise, we can calculate $g \cdot h - f$ and see if that is 0 or not. Thus, if an element of Z has finitely many divisors, this method can be used to look for divisors on $f(X)$.

Eisenstein's criterion

Given that p is a prime in Z and $f(X)$ is monic in $Z[X]$ such that $f \bmod p$ is of the form uX^d where u is a unit in $Z/\langle p \rangle$.

If $f(X) = g(X)h(X)$ is a factoring in $Z[X]$, then g and h are monic. Now, we get $uX^d = (g \bmod p)(h \bmod p)$ which shows that $g = vX^m$ and $h = wX^n$ where $m + n = d$.

It follows that $g(0)$ and $h(0)$ are multiples of p . From this it follows that $f(0) = g(0)h(0)$ is in $\langle p^2 \rangle$.

Turning this around, we obtain the following.

If p is a prime in Z , $f(X)$ is monic in $Z[X]$ such that $f \bmod p$ is of the form uX^d where u is a unit in $Z/\langle p \rangle$, and $f(0)$ is not in $\langle p^2 \rangle$, then f is irreducible.